Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Emergency Management of Hyperkalaemia in Adults

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SUMMARY

This guideline outlines the process for emergency management of adult patients with hyperkalaemia (high potassium levels). It covers identification, causes, assessment and treatment of hyperkalaemia via both oral and intravenous routes.

KEY POINTS

Hyperkalaemia

↓

Consider repeat blood test to exclude artefactual hyperkalaemia

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Decision to Treat

↓

Identify Cause(s)

↓

Clinical Assessment

↓

TREATMENT and GOALS

↓

1. Protect the Heart
CALCIUM GLUCONATE IV

↓

2. Shift Potassium into Cells
INSULIN ACTRAPID & GLUCOSE - SALBUTAMOL NEBS - SODIUM BICARBONATE IV

↓

3. Remove Potassium from the Body
SODIUM ZIRCONIUM (oral) - HAEMODIALYSIS/FILTRATION

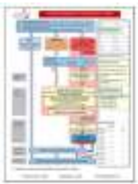
↓

4. Monitoring
ECG - Potassium - Glucose

↓

5. Prevent Recurrence

Also refer to treatment summary algorithm (UK Kidney Association) in appendix 3:



CONTENTS

1 INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 Hyperkalaemia is a common medical emergency and can be potentially life threatening. The major consequences of hyperkalaemia are arrhythmias and cardiac arrest therefore early recognition and timely management is essential.

1.3 The urgency to treat hyperkalaemia depends on the presence of symptoms and signs, how elevated serum potassium is and the aetiology of hyperkalaemia.

2 BACKGROUND

2.1 This guidance replaces a pair of historical legacy guidelines in place at RDEFT and NDHT prior to integration.

3 DEFINITIONS

3.1 Hyperkalaemia is defined as a serum potassium of ≥5.5mmol/L (European Resuscitation Council (ERC) Guideline definition).

3.2 Classification according to severity:

Severity	Blood Concentration (mmol/L)
Mild	5.5 - 5.9
Moderate	6.0 - 6.4
Severe	≥ 6.5

4 DECISION TO TREAT

4.1 Consider hyperkalaemia to require urgent treatment if:

- There are characteristic ECG changes.

OR

- There are symptoms (muscle weakness or flaccid paralysis, palpitations, paraesthesia).

OR

- Serum potassium >6.0 mmol/L

OR

- Patient is in cardiac arrest

4.2 If the above features occur at **ANY** level of serum potassium ≥ 5.5mmol/l, then consider it severe.

4.3 If mild hyperkalaemia (serum level 5.5-5.9 mmol/L), and not having any of the above features, consider a repeat blood test before initiating treatment to exclude artefactual hyperkalaemia. Mild hyperkalaemia is often well tolerated and commonly encountered in patients with chronic kidney disease.

5 CAUSES

5.1 Comprehensive history, examination and review of medication history can help determine the cause of hyperkalaemia.

5.2 Excessive potassium in plasma can be caused by:

- Excessive intake
- Reduced renal excretion
- A shift in potassium from intra to extra-cellular space
- Technical errors causing pseudo-hyperkalaemia

5.3 There is a multitude of causes and sometimes the aetiology may be multifactorial. For drugs frequently implicated in hyperkalaemia, please refer to appendix 1 and for non-pharmacological causes, please refer to appendix 2. Risk factors to consider:

- Acute Kidney Injury
- Dialysis dependency (hemodialysis or peritoneal dialysis)
- Chronic Kidney disease stages four and five (eGFR < 30 mL/min)
- Drugs
- Cardiac failure
- Diabetes Mellitus (renin-angiotensin drugs, diabetic ketoacidosis)
- Liver disease (spironolactone, hepato-renal failure)
- Addison's disease (primary adrenal insufficiency)
- Hyporeninaemic hypoaldosteronism (Type IV renal tubular acidosis)

6 ASSESSMENT

6.1 Start with assessment of the patient (ABCDE) and National Early Warning Score (NEWS).

6.2 Send an arterial or venous blood sample using a point-of-care blood gas analyser whilst awaiting the results from a formal laboratory measurement. Blood gas analysers provide a rapid estimation of serum potassium levels to guide management but don't replace the need for formal laboratory testing (e.g. haemolysis is not considered on a blood gas where as it is on a formal laboratory sample).

6.3 Perform a 12 lead ECG if Serum potassium > 5.5 mmol/L.

6.4 Assess the risk of arrhythmia, including potential rate of rise in serum potassium (e.g. rhabdomyolysis or oliguric renal failure).

6.5 Renal dialysis patients should be discussed directly with the renal team (based at the Wonford site) – contacted via switchboard.

6.6 Continuous ECG monitoring (three lead) if patient has adverse feature on ECG, Potassium >6.5 mmol/l or K>6.0 and patient unwell or rapidly rising potassium. This may require transfer to a higher care area.

6.7 If a patient suffers a cardiac arrest at any point during their management of hyperkalaemia, they should be managed as per the Resuscitation Council's Advanced Life Support guidelines.

7 EMERGENCY MANAGEMENT

STEP 1 – PROTECT THE HEART

7.1 **Intravenous Calcium Salt**:

If there are **ECG changes** (peaked T waves, absent or flattened P waves, progressive lengthening of the PR interval and QRS duration, Sine wave, bradycardia, ventricular tachycardia) then **administer ONE of the following**:

FIRST LINE	**30mLs of 10% calcium gluconate IV bolus over** 10 minutes
SECOND LINE (when 1st line not available)	**10mLs of 10% calcium chloride IV bolus over** 5 minutes

- USE LARGE VEIN (for both calcium formulations which can be given undiluted).
- Repeat dose after 5 minutes if ECG changes persist.
- Do not administer sodium bicarbonate through the same access or mix with other drugs.
- Onset of action is within 3 minutes, and duration of action is between 30 - 60 minutes.

Note
- 10 mL x 10% *calcium gluconate* = 2.26 mmol Ca^{+2}.
- 10 mL x 10% *calcium chloride* = 6.80 mmol Ca^{+2}.

- Adverse effects that have been reported from administration of IV calcium salts include:
    - Erythema, tenderness and in the worst cases, tissue necrosis if extravasation occurs (calcium gluconate being less irritant to the tissues than the calcium chloride);
    - Hypotension;
    - Peripheral vasodilation;
    - Hot flushes (following rapid infusion); and
    - Arrhythmias.

7.2 Patients must have continuous cardiac monitoring to assess their heart rhythm during treatment. This will usually be via a three lead ECG monitor.

7.3 If the patient is taking Digoxin: the calcium gluconate should be given slowly (mixed with 100mLs 5% glucose and given over 20 minutes) as rapid calcium administration may precipitate myocardial digoxin toxicity.

7.4 Remember to repeat ECG after calcium has been given to look for narrowing of QRS complexes, increased heart rate or resolution of arrhythmia.

7.5 In cardiac arrests where hyperkalaemia is felt to be contributory, hyperkalaemia should be confirmed using a blood gas analyser if available and IV calcium salts should be given (*calcium chloride* preferred as can be administered more rapidly). This can be repeated if cardiac arrest is refractory or prolonged.

8 CONTINUED MANAGEMENT

STEP 2 - SHIFT POTASSIUM INTO CELLS/OUT OF THE BODY

8.1 **Insulin (Actrapid) and Glucose** (off license indication)**:**

- If there is access to a **large vein**, then administer 10 units of insulin Actrapid in 50 mL of 50% glucose given over 15-30 minutes**.**
- If only **peripheral line** available, give 10 units insulin Actrapid in 125 mL 20% glucose over 5-15 minutes.

Or

- If there is access to a **large vein**, then administer 10 units of insulin Actrapid in 50mL of 50% dextrose given over 15-30 minutes**.**
- Effects peak at 30-60 min & last for up to 6 hours.
- Patient is at risk of hypoglycaemia following insulin-glucose treatment therefore regular glucose monitoring is needed. Refer to section 6.17 for recommendations.
- Follow diabetic ketoacidosis (DKA) protocol regarding insulin and glucose for patients with DKA.
- In cardiac arrests where hyperkalaemia is felt to be contributory, the insulin and dextrose may be given by rapid injection.

8.2 **Nebulised Salbutamol** (off license indication):

- Administer 10 mg – 20 mg nebulised salbutamol (10 mg in patients with IHD, severe tachycardia).
- For nebulisation, dilute nebuliser solution with a suitable volume of sterile Sodium Chloride 0.9% solution according to nebuliser type and duration of administration.
- In combination with insulin/glucose, salbutamol can lower serum potassium by an additional 0.5-1mmol/L.
- Onset of action 30mins, effect lasts up to 2 hours.
- Salbutamol should not be used as monotherapy in the treatment of severe hyperkalaemia.
- Nebulised salbutamol may not be effective, especially in those patients taking beta-blockers or digoxin, but may be used as first-line treatment whilst IV access is being established.

8.3 **Sodium bicarbonate**:

- Its use should be limited to cases of severe metabolic acidosis. It is sometimes used in cardiac arrest situations under specialist guidance.

  Refer to intensive care and/or renal team in cases with severe metabolic acidosis for advice on how to approach and manage.

- Can shift potassium from the extracellular to intracellular by increasing blood PH. The evidence for routine use in hyperkalaemia is controversial, its effects are of variable onset and not sustained.

STEP 3 - REMOVE POTASSIUM FROM THE BODY

8.4 Sodium zirconium cyclosilicate (*Lokelma*®)

- Indications for use: (as per NICE guidance 2022):
  - Acute life-threatening hyperkalaemia alongside standard care.
  - Persistent hyperkalaemia in chronic kidney disease 3b-5 or heart failure if potassium >/= 6 mmol/L, patient not on haemodialysis, or to enable them to be optimised with a RAASi (renin angiotensin-aldosterone inhibitor).
- The recommended starting dose of Lokelma is 10 g, administered three times a day for up to 72 hours. It is taken orally as a suspension in water.
- Recommended maintenance dose is 5g daily but that should only be prescribed if requested by the renal team.
- If normokalaemia is not achieved after 72 hours of treatment, discuss with the renal team as likely other treatment approaches should be considered (it is usually expected to achieve normokalaemia within 24 to 48 hours).
- Cautions – can cause QT interval lengthening as a result of a reduction in serum potassium. May be opaque to x-rays which needs to be considered if having abdominal X-rays. It has minimal side effects which include hypokalaemia, oedema and gastrointestinal disturbance.
- There is a similar drug (Patiromer) which is currently not used in this trust for the management of hyperkalaemia (but is mentioned in the Algorithm in Appendix 3).

8.5 Calcium Resonium

- Do **not** use cation-exchange resins (e.g. calcium resonium) for the emergency treatment of severe hyperkalaemia**.** The onset of action is slow (up to five days) and they can cause severe gastrointestinal side-effects. Usually given orally 3-4 times a day. Can be given rectally once daily if oral route is unavailable. There is no role for calcium resonium in the management of hyperkalaemia if Lokelma is available.

8.6 Haemodialysis or Haemofiltration

- These are the most effective but invasive treatments for severe resistant hyperkalaemia where other measures have been unsuccessful or if there is ongoing tissue damage. **Critical Care and the renal team should be consulted early in these circumstances.**

STEP 4 - MONITORING

8.7 **ECG:** Perform continuous ECG monitoring with frequent EWS during treatment.

8.8 **Monitoring Potassium:** Close monitoring of potassium is essential to assess response to management and to identify rebound after initial management wears off.

- Measure serum potassium at least 1, 2, 4, 6 and 24 hours after identification and treatment of moderate or severe hyperkalaemia.
- Aim to reduce serum potassium < 6.0 mmol/L.

8.9 **Monitoring Glucose:** Monitor blood glucose at regular intervals up to 12 hours after insulin-glucose infusion for treating hyperkalaemia to pick up hypoglycaemia early and treat.

- Monitor capillary glucose at 0, 15, 30 minutes after starting treatment and then hourly up to 6 hours.
- A continuous glucose infusion can be instituted if ongoing hypoglycaemia.
- If pre-treatment blood glucose < 7.0 mmol/L, an infusion of 10% glucose at 50 mL/hour for 5 hours (25 g) can be initiated following insulin-glucose treatment to avoid hypoglycaemia with ongoing monitoring of glucose levels aiming at a target blood glucose 4-7 mmol/L.
- Delayed hypoglycaemia is commonly reported with insulin/glucose management.
- The risk of severe hypoglycaemia for patients with acute kidney injury or end-stage renal disease is heightened in patients with lower body weight and creatinine clearance.

STEP 5 - PREVENT RECURRENCE

- Stop offending medications, it may not be appropriate to stop all.
- Those most commonly implicated are as follows: Potassium supplements, salt substitutes (e.g. LoSalt), ACE-I, Angiotensin II Receptor Blockers, NSAIDs, spironolactone, eplerenone, amiloride, trimethoprim, digoxin. Refer to Appendix 2.
- Once urgent treatment has been instigated ensure patient is placed on a low potassium diet. It is reasonable to refer patients with CKD stage 4 and 5 to dietician prior to discharge (can be done via EPIC order).
- Management of hyperkalaemia and continuing care relates to the disease process that led to hyperkalaemia therefore determining the underlying aetiology is essential. Please refer to appendix 3.

9 MONITORING COMPLIANCE WITH THIS GUIDELINE

9.1 Standards/Key Performance Indicators

Key performance indicators comprise:

- Number of patients with hyperkalaemia duration of over 24 hrs
- Complications associated with hyperkalaemia
- Internal auditing

9.2 Process for implementation and monitoring compliance and effectiveness

- The policy will be distributed to all ward managers & medical staff for implementation.
- Incident reporting (Datix) speciality governance review to monitor compliance.
- Medication Safety Group Quarterly reports to identify any spikes in reporting where relates to IV calcium products.
- QI audit project review prescribing of injectable calcium after 3-6 months to check compliance & provide assurance on risk.

9.3 Non-compliance will be reported on Datix.

10 ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

Medicines Management Policy (2022)

11 REFERENCES

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- BNF online. https://bnf.nice.org.uk/drugs/sodium-zirconium-cyclosilicate/
- National Patient Safety Alert: Potential risk of under-dosing with calcium gluconate in severe hyperkalaemia. MHRA 27-Jun-23. https://www.gov.uk/drug-device-alerts/national-patient-safety-alert-potential-risk-of-underdosing-with-calcium-gluconate-in-severe-hyperkalaemia-natpsa-slash-2023-slash-007-slash-mhra
- Medusa Injectable Medicines Guide online via HUB. https://www.medusaimg.nhs.uk/?ID=f420cd6252f971a5f3057ceaa89a527a981

APPENDIX 1: DRUGS ASSOCIATED WITH CAUSING HYPERKALAEMIA

Decreased Excretion of Potassium

- ACE inhibitors
- ARBs
- NSAIDs
- Spironolactone
- Eplerenone
- Ketoconazole
- Heparin/LMWH
- Trimethoprim
- Pentamidine

Redistribution (Increased extracellular shift)

- Beta-blockers
- Digoxin
- Suxamethonium
- Mannitol

Excess potassium

- Potassium supplements
- Potassium containing laxatives e.g. Movicol, Laxido
- LoSalt
- Excess diet (usually only if concurrent renal impairment)

Other Causes

- Amphotericin
- Nicorandil
- Propofol
- Zoledronic acid

There are rarer associations between medications and hyperkalaemia and it is important every prescribed medication is reviewed in the context of hyperkalaemia.

APPENDIX 2: NON-PHARMACEUTICAL CAUSES OF HYPERKALAEMIA

Renal

- Acute kidney injury
- Chronic kidney disease (especially ESRD on dialysis)
- Interstitial nephritis or tubular disease

Lack of aldosterone

- Addison's disease
- Renal Tubular Acidosis T4
- Rarely congenital adrenal hyperplasia
- Advanced CCF

Redistribution

- Acidosis e.g. Diabetic ketoacidosis
- Cell tissue breakdown, e.g. trauma, crush injury, rhabdomyolysis, haemolysis, tumour lysis, transfusion

Pseudo-hyperkalaemia

- Long transit time or poor storage
- Difficult venepuncture
- Wrong order of drawing blood tubes
- Haematological conditions e.g. high platelets
- Haemolysed sample

APPENDIX 3: EMERGENCY TREATMENT ALGORITHM

(Adopted from UK Kidney Association with amendments made to align with Trust policy)